

Chendong Wang

✉ cdwang96@gmail.com | ☎ +1-763-501-1816 | 🌐 [cedricbeta](https://cedricbeta.com) | 🌐 cdwang96.github.io | 🌐 [chendong-wang-aaa9b3b8](https://chendong-wang-aaa9b3b8.github.io)

Research Focus

- **Training and Inference Systems for Large-scale AI:** Building scalable infrastructure for distributed training, RL-based post-training, rollout/generation, and high-throughput long-context inference.
- **Systems-Agent Co-design:** Enhancing long-horizon agent reasoning by bridging inference engines and agentic logic through semantic memory management, KV-cache optimization, and cache-aware execution.

Education

University of Wisconsin-Madison

Ph.D. in Computer Science

2021/09 — Now

Madison, WI, USA

University of Minnesota

M.S. in Computer Science, GPA 3.926/4.0

2019/09 — 2021/06

Minneapolis, MN, USA

Fudan University

B.S. in Theoretical and Applied Mechanics, Top 20%; Minor in Data Science

2014/09 — 2019/07

Shanghai, China

Selected Publications

Video-in-the-Loop: Span-Grounded Long Video QA with Interleaved Reasoning

ICML 2026

Chendong Wang, Donglin Bai, Yifan Yang, Xiao Jin, Anlan Zhang, Rui Wang, Shiqi Jiang, Yuqing Yang, Hao Wu, Qi Dai, Chong Luo, Ting Cao, Lili Qiu, Suman Banerjee

Zoomer: Enhancing MLLM Performance with Adaptive Image Focus Optimization

TMLR

Jiaxu Qian, Chendong Wang, Yifan Yang, Chaoyun Zhang, Huiqiang Jiang, Xufang Luo, Yu Kang, Qingwei Lin, Anlan Zhang, Shiqi Jiang, Ting Cao, Tianjun Mao, Suman Banerjee, Guyue Liu, Saravan Rajmohan, Dongmei Zhang, Yuqing Yang, Qi Zhang, Lili Qiu

Switched or Switchless: An Empirical Study of SAN Architecture for Disaggregated Storage

NSDI 2026

Chendong Wang, Joontaek Oh, and Ming Liu

VoLUT: Efficient Volumetric Streaming Enhanced by LUT-based Super-resolution

MLSys 2025

Chendong Wang, Yifan Yang, Lili Qiu, Anlan Zhang, Yuqing Yang, Xinyang Jiang, Suman Banerjee

Habitus: Boosting Mobile Immersive Content Delivery through Full-body Pose Tracking and Multipath Networking

NSDI 2024

Anlan Zhang, Chendong Wang, Yumin Hu, Ahmad Hassan, Zejun Zhang, Bo Han, Feng Qian, Shichang Xu

YuZu: Super-resolution Enhanced Volumetric Video Streaming

NSDI 2022

Anlan Zhang, Chendong Wang, Bo Han, Feng Qian

Efficient Volumetric Video Streaming Through Super Resolution

HotMobile 2021

Anlan Zhang, Chendong Wang, Bo Han, Feng Qian

Mobile Volumetric Video Streaming Enhanced by Super Resolution

MobiSys 2020 Poster

Anlan Zhang, Chendong Wang, Xing Liu, Bo Han, Feng Qian

Firefly: Untethered Multi-user VR for Commodity Mobile Devices

USENIX ATC 2020

Xing Liu, Christina Vlachou, Feng Qian, Chendong Wang, and Kyu-Han Kim

Work Experience

Fireworks AI

2026/09 — Now

MTS, Training

San Mateo, CA, USA

- Working on training systems for large-scale AI workloads.

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Google

2026/05 — 2026/09

SWE PhD Intern

Sunnyvale, CA, USA

- Engineered a straggler injection and analysis framework within the NCCL FasTrak network plugin across A3 Ultra and A4 GPU clusters, enabling controlled emulation of slow nodes to evaluate large-scale collective communication.
- Characterized straggler propagation in multi-node collective communication workloads to root-cause performance issues, tracing cross-GPU slowdowns to shared host-side transport resources and distinguishing software-path head-of-line blocking from hardware topology effects.
- Developed profiling and diagnosis workflows combining hot-path logging, flow-group analysis, and NIC/NUMA topology inspection to improve observability of intra/inter-node traffic and localize communication bottlenecks in distributed ML training.

Meta Inc.

2025/05 — 2025/08

SDE Intern, Mentor: Jun Seok Lee

Bellevue, WA, USA

- Developed production ML pipelines for ads retrieval and accelerated early-stage retrieval by **2.5x** with no quality degradation.
- Built an LLM evaluation pipeline for automated retrieval-quality assessment, regression tracking, and model behavior analysis.
- Designed scalable evaluation workflows to compare model outputs, detect regressions, and support rapid iteration on ML systems.

ByteDance Inc.

2024/05 — 2024/08

Research Intern, Mentor: Yan Sun

San Jose, CA, USA

- Developed a topology-aware collective communication optimization framework for PCIe GPU clusters, improving P2P bandwidth by up to **30%** through shared-memory transfer and chunk-size tuning.
- Investigated RDMA scalability bottlenecks for expert-parallel LLM inference and implemented bitmap-based queue-pair pooling to efficiently manage over **1000** queue pairs.
- Profiled GPU communication and serving workloads to identify system-level bottlenecks in distributed LLM inference.

Microsoft Research Asia (MSRA)

2023/05 — 2023/08

Research Intern, Mentor: Prof. Lili Qiu

Shanghai, China

- Improved volumetric-video inference with LUT-based acceleration, delivering over **10x** frame-rate speedup.
- Studied 3D point-cloud upsampling methods and drafted a conference submission on efficient spatial enhancement.

Hewlett-Packard Company (HP)

2018/10 — 2019/05

R&D Intern

Shanghai, China

- Implemented a text-to-speech prototype using WaveRNN-based neural speech synthesis.
- Researched NLP methods for reasoning-oriented language understanding.

Honors & Awards

Third Prize, China Undergraduate Mathematical Contest in Modeling

2017

Third Prize, Elite Experiment Scholarship

2017, 2018

Skills

- **Programming:** Python, C++, C, Matlab
- **ML Systems:** Distributed training systems, RL post-training infrastructure, rollout/inference pipelines, FSDP, GPU communication, RDMA, expert-parallel inference
- **LLM Systems:** Long-context inference, high-throughput serving, KV-cache optimization, scalable evaluation pipelines, memory-efficient inference
- **Agentic Systems:** Semantic memory eviction, context-rot mitigation, cache-aware planning, long-horizon software engineering agents, graph-based KV memory management
- **Systems:** Performance profiling, collective communication, GPU cluster communication, storage systems, bottleneck diagnosis

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